Federal Economist Compensation & PhD Placement: Evidence by Gender

Updated Data and New Sources

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Outline

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2. Federal Salary Analysis
3. Comparison to Prior Work
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Overview

This project extends prior work on gender equity among economists in two ways:

1. Updates the federal salary analysis through FY2024 using FedsDataCenter (previously through FY2015 in Foster, Manzella, McEntarfer & Sandler 2020)
2. Introduces a new data source on PhD economist initial placements (44 programs, 1991–2025) to complement the LEHD-based analysis in Foster, McEntarfer & Sandler (2023)

Key questions:

- How has the federal economist gender pay gap evolved since 2015?
- How do initial placement shares compare across sectors and over time?
- How does the gender composition of placements compare to the full SED/LEHD sample?

Data Sources

Federal salary data

- Source: FedsDataCenter.com (OPM data via FOIA)
- Years: FY2015–FY2024
- ≈4,300–4,700 economists per year
- Occupation code: "Economist"
- Fields: name, agency, grade, pay plan, salary, bonus
- Gender assigned via `gender_guesser` (offline, ≈84% coverage)

PhD placement data

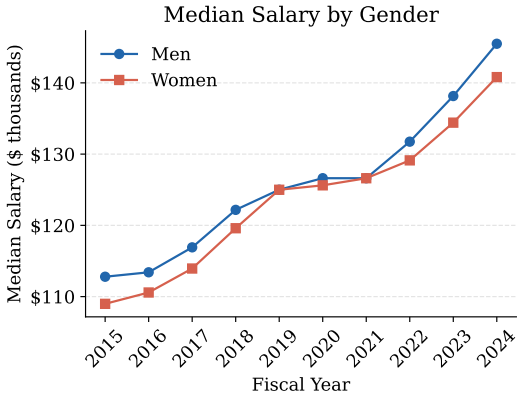
- Source: econphdplacements.com
- 8,824 placements from 44 programs
- Includes US and international programs (e.g., Stockholm, UBC, LSE)
- Years: 1991–2025
- Fields: name, year, program, field, placement, category
- Gender assigned via `gender_guesser` (≈51% coverage)

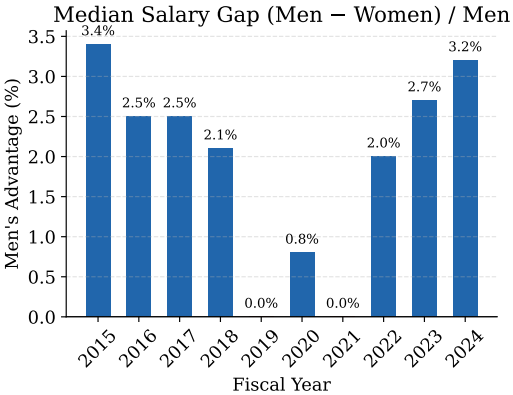
A Note on Gender Assignment

Both datasets use **name-based gender assignment** via `gender_guesser` (an offline library covering ≈40,000 names):

- Returns: *male*, *mostly_male*, *female*, *mostly_female*, *androgynous*, *unknown*
- We assign male/female from the first two; leave the rest unassigned
- Salary data: **84%** coverage (OPM names are mostly Anglo-American)
- Placement data: **51%** coverage (many international names not in library)
- Unassigned names skew international and likely male-heavy ⇒ female shares in placement data are *understated*
- Prior papers (Foster et al. 2020, 2023) use **self-reported** SED gender (100% coverage, no bias)

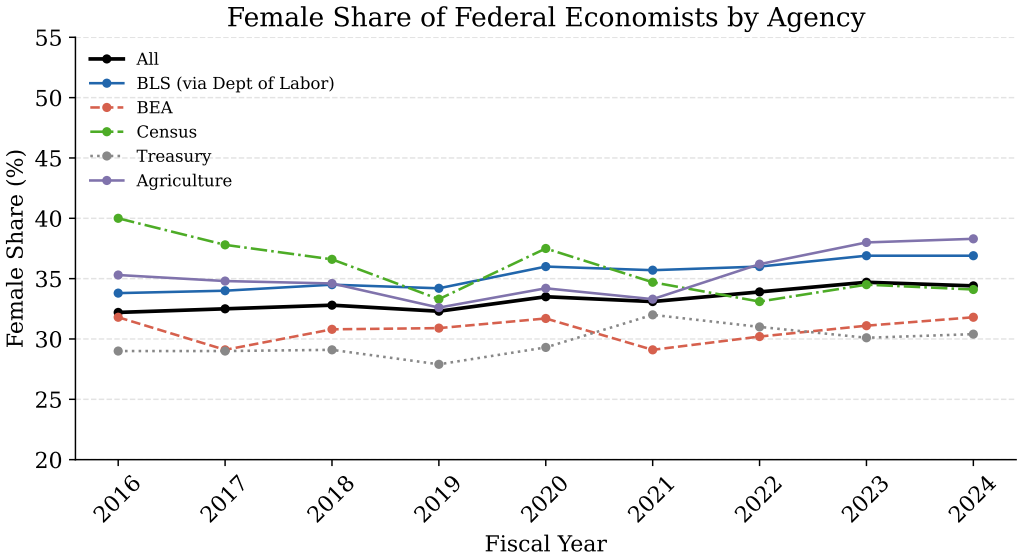
Median Salary by Gender, FY2015–2024



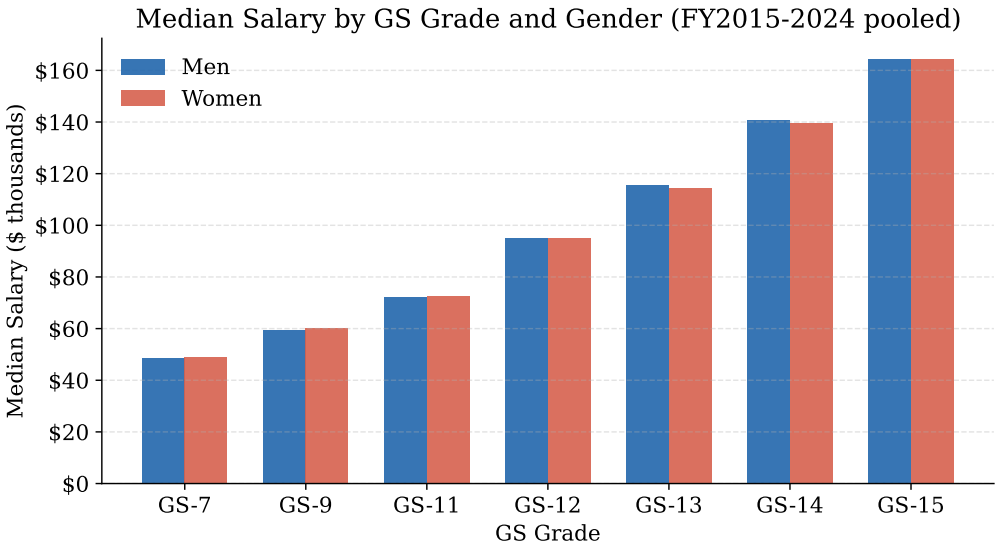


- Gap peaked at 3.4% in 2015, nearly closed in 2019–2021, then **re-widened to 3.2%** by 2024
- 2024 medians: men $145,489 women $140,807

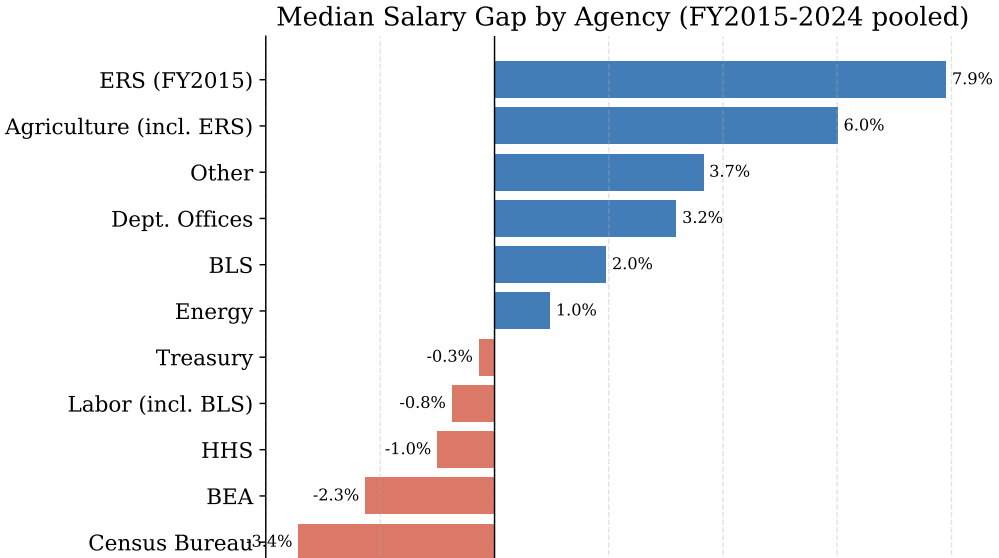
Female Share of Federal Economists by Agency



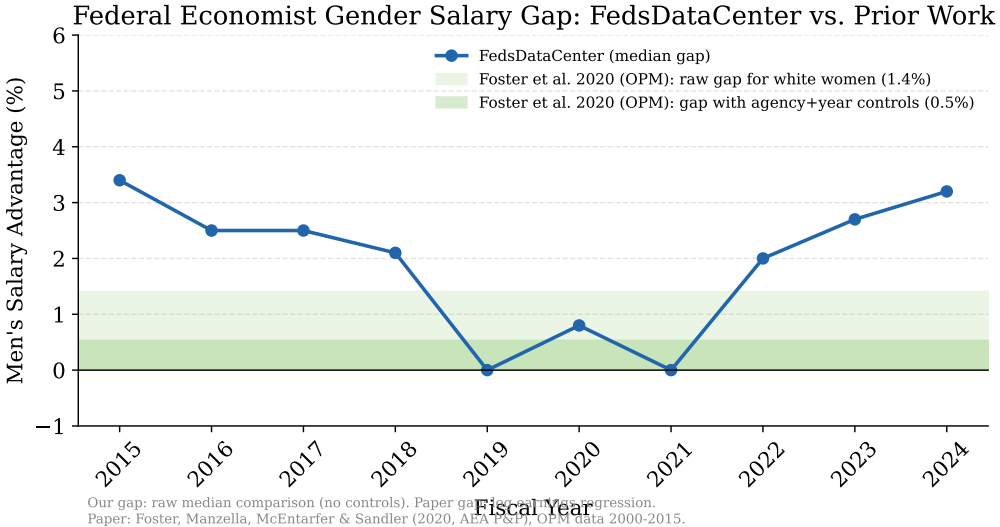
Is the Gap Within or Between Grade?



Salary Gap by Agency



Federal Salary Gap: FedsDataCenter vs. Foster et al. (2020)



Our gap: raw median comparison (no controls). Paper gap: agency controls regression.
Paper: Foster, Manzella, McEntarfer & Sandler (2020, AEA P&P), OPM data 2000-2015.

Federal Salary Gap: Comparison Summary

Source	Sample	Specification	Gap
FedsDataCenter (FY2015)	All GS economists	Raw median	3.4%
FedsDataCenter (FY2024)	All GS economists	Raw median	3.2%
FedsDataCenter (FY2019–21)	All GS economists	Raw median	≈0%
Foster et al. (2020)	PhD economists, OPM	Raw log gap (white women)	1.4%
Foster et al. (2020)	PhD economists, OPM	Agency + year FE	0.5%
Foster et al. (2023)	PhD economists, LEHD	Raw log gap, government	8.6%
Foster et al. (2023)	PhD economists, LEHD	Full controls, government	3.8%

- Our raw 3% median gap vs. OPM-based 1.4% raw log gap: broadly consistent
- LEHD-based 8.6% is larger because it is **PhD-only** and captures broader compensation (not just base salary)

PhD Economist Placement Data: Overview

Sample characteristics

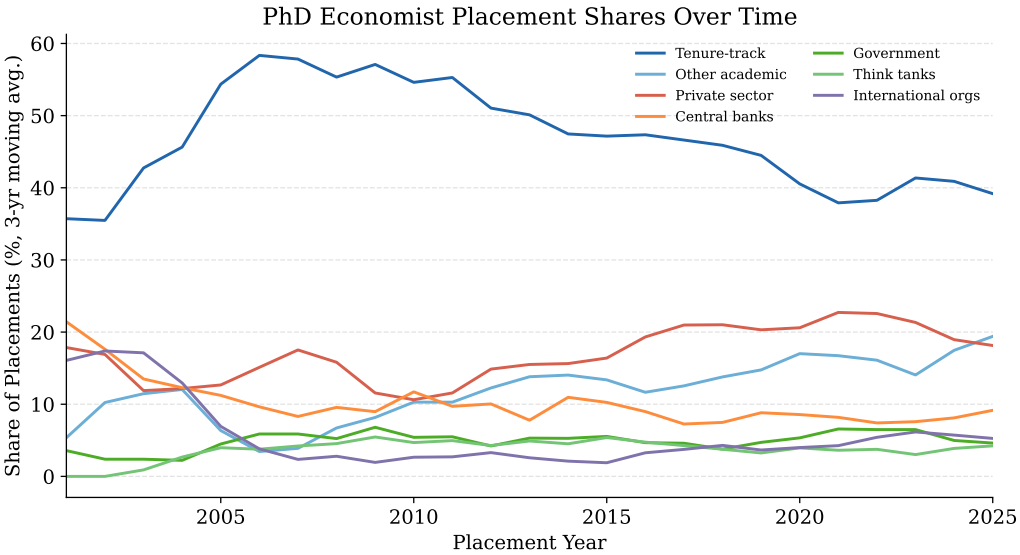
- 8,824 placements, 44 programs
- Placement years: 1991–2025
- Includes international programs: Stockholm, UBC, LSE, Sciences Po, Mannheim, EUI, CEMFI, ...
- 7 placement categories: tenure-track, other academic, private sector, central banks, government, think tanks, international orgs

Overall placement distribution

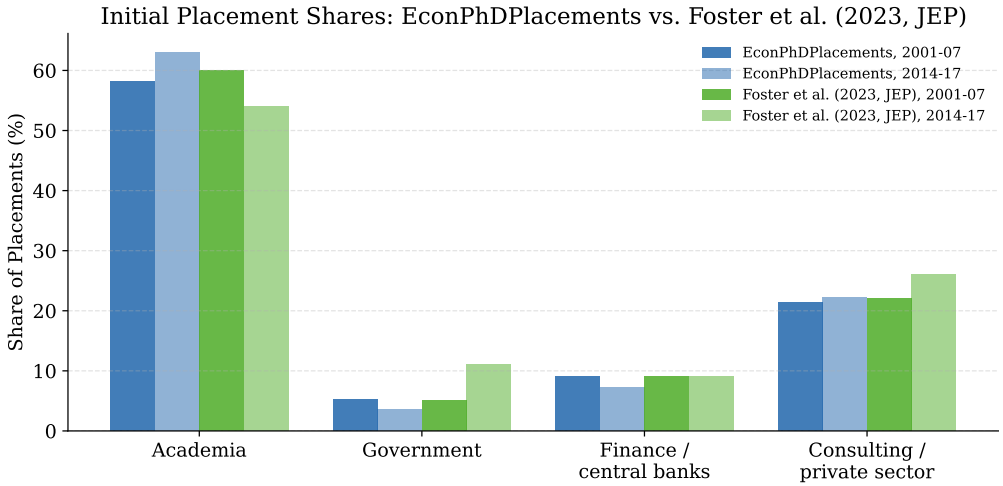
Category	Share
Tenure-track	49.4%
Private sector	18.8%
Other academic	12.2%
Central banks	7.2%
International orgs	4.6%
Think tanks	3.8%
Government	4.1%

Academia total: 61.6%

Placement Trends Over Time



Placement Shares: EconPhDPlacements vs. Foster et al. (2023, JEP)



Note: 'International orgs' (4-6% in EconPhDPlacements) not captured in paper (LEHD = US only).
Paper: Foster, McEntarfer & Sandler (2023, JEP). Govt in paper = NAICS 92 only.

Reconciling the Differences

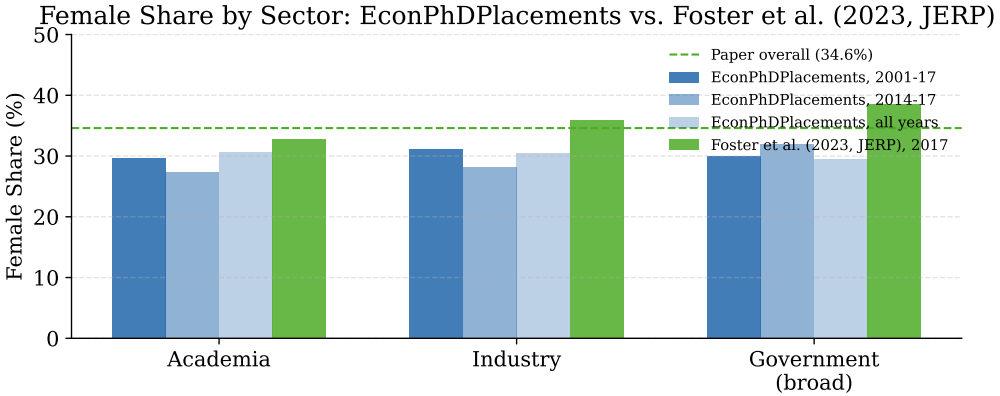
Our academia share (63%, 2014–17) is higher than paper's (54%, all programs):

- Our 44 programs skew toward **top research universities** — even for top-5 programs, paper shows 58.6%
- Placement website data = **announced** initial placements; LEHD = actual employment (captures people who planned academia but moved)
- International programs in our sample (Stockholm, UBC, . . .) may have higher academic placement rates

Our government share (3.6%, 2014–17) is lower than paper's (11%):

- Top programs go to government less (paper's top-5: 8.6%; others: 11.4%)
- Government placements may be under-reported on department websites
- Paper's government includes all levels (federal + state/local, NAICS 92)

Female Share by Sector: EconPhDPlacements vs. Foster et al.



Note: Gender assigned via gender_guesser (50.5% of placements). Unassigned names skew international (likely male-heavy), so our female shares are likely understated. Paper uses self-reported SED gender (100% coverage). D&E = Diversity & Equity in Labor Market Outcomes for Economists (Foster, McEntarfer & Sandler, 2023, JERP).

Female Share in Placements: Caveats and Takeaways

Our estimates (29–32%) are below the paper's (33–38%):

- **Gender assignment gap:** only 51% assigned; unassigned names are disproportionately international and likely male-heavy $\Rightarrow$ our female shares are understated by an unknown amount
- **Initial placements vs. cross-section:** paper shows the 2017 stock of employed PhD economists (includes earlier cohorts with higher female retention); ours shows the flow of new placements
- **Ranking within sectors is preserved:** government $>$ industry $>$ academia in both datasets

Next step: use gender-specific name databases (e.g., Social Security Administration data, genderize.io) to improve coverage for international names

Summary

Federal salary (FedsDataCenter, FY2015–2024):

- Raw median gender gap: 0–3.4%, broadly consistent with Foster et al. (2020)
- Gap is entirely between-grade, not within-grade; BEA, Census, BLS show female salary *advantage* at median
- Female share rising: 32% $\rightarrow$ 34%; gap re-widened post-2021

PhD placements (econphdplacements.com, 1991–2025):

- Sample skews toward top programs; academia share higher than national average
- Government placements under-represented vs. LEHD-based paper
- International orgs (4–6%) visible here but absent from LEHD data
- Female share understated due to gender assignment coverage; ordering across sectors preserved

Next steps:

- Improve gender assignment for international names
- Link placement and salary data on individual economists
- Extend to regression-based gap estimates controlling for grade/agency

References I

- Foster, L., Manzella, J., McEntarfer, E., & Sandler, D.H. (2020). "Employment and Earnings for Federal Government Economists: Empirical Evidence by Gender and Race." *AEA Papers and Proceedings*, 110, 210–214.
- Foster, L., McEntarfer, E., & Sandler, D.H. (2023). "Diversity and Equity in Labor Market Outcomes for Economists." *Journal of Economics, Race, and Policy*.
- Foster, L., McEntarfer, E., & Sandler, D.H. (2023). "Early Career Paths of Economists Inside and Outside of Academia." *Journal of Economic Perspectives*, 37(4), 231–250.
- Data: FedsDataCenter.com (OPM federal salary data, FY2015–2024)
- Data: econphdplacements.com (PhD economist placement data, 1991–2025)